

Dashboard

 Search patients by name or ID...



8

TOTAL RECORDS



8

WAITING ROOM



0

IN TREATMENT



4

SEVERE / CRITICAL

NEXT UP FOR TREATMENT

[View Full Queue](#)



Jane Smith P002

⌚ Waiting: 1h ago

LEVEL 5 - CRITICAL



Fatima Hassan P008

⌚ Waiting: 1h ago

LEVEL 5 - CRITICAL



Bob Davis P004

⌚ Waiting: 1h ago

LEVEL 4 - SEVERE



Diana Lee P006

⌚ Waiting: 1h ago

LEVEL 4 - SEVERE

QUICK ACTIONS



Register Patient



Triage



Medical Staff



Analytics

Patient Records

 Search patients by name or ID...



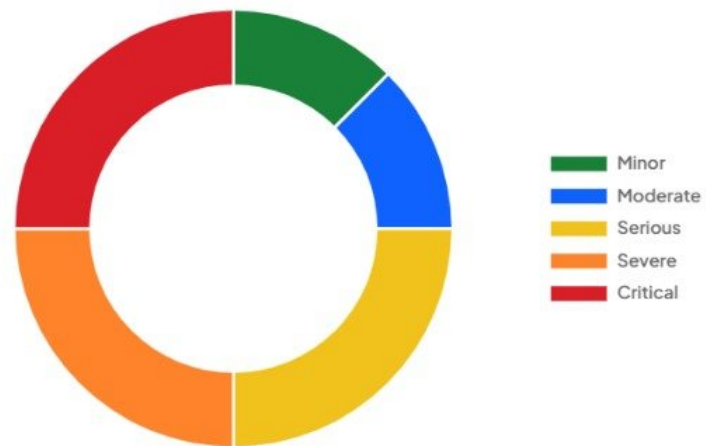
ID	PATIENT NAME	SEVERITY	STATUS	ARRIVED	ACTIONS
P001	John Doe	LEVEL 3 - SERIOUS	WAITING	Sep 1, 9:36 PM	
P002	Jane Smith	LEVEL 5 - CRITICAL	WAITING	Sep 1, 9:36 PM	
P003	Alice Brown	LEVEL 3 - SERIOUS	WAITING	Sep 1, 9:36 PM	
P004	Bob Davis	LEVEL 4 - SEVERE	WAITING	Sep 1, 9:36 PM	
P005	Carlos Garcia	LEVEL 2 - MODERATE	WAITING	Sep 1, 9:36 PM	
P006	Diana Lee	LEVEL 4 - SEVERE	WAITING	Sep 1, 9:36 PM	
P007	Ethan Wright	LEVEL 1 - MINOR	WAITING	Sep 1, 9:36 PM	
P008	Fatima Hassan	LEVEL 5 - CRITICAL	WAITING	Sep 1, 9:36 PM	

Analytics

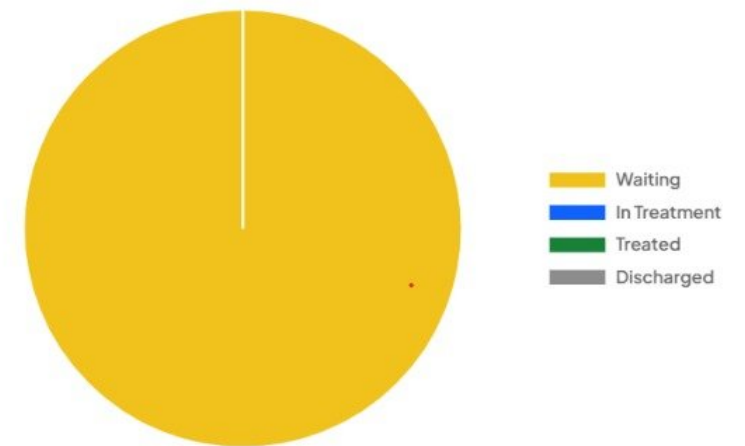
Q Search patients by name or ID...



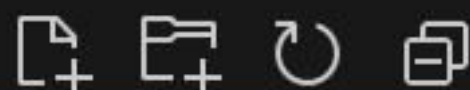
QUEUE BY SEVERITY



PATIENT STATUS OVERVIEW



✓ janani datastructure



>  .vercel

>  app

 .gitignore

U

 emergency_system.py

 index.html

 presentation.html

 Project_Report.md

 README.md

 vercel.json

</> presentation.html 2 X

</> presentation.html > ...

```
2   <html lang="en">
3   <head>
8     <style>
97     .btn {
100       font-weight: bold;
104       text-decoration: none;
105       cursor: pointer;
106       transition: all 0.3s ease;
107       border: none;
108       font-size: 1rem;
109       min-height: 44px;
110       min-width: 44px;
111     }
112
113     .btn-primary {
114       background: var(--gradient);
115       color: white;
116       box-shadow: 0 4px 15px rgba(15, 118, 110, 0.4);
117     }
118
119     .btn-primary:hover {
120       box-shadow: 0 6px 20px rgba(15, 118, 110, 0.6);
121       transform: translateY(-2px);
122     }
123
124     .btn-danger {
125       background: var(--accent);
126       color: white;
127     }
128
129     .btn-danger:hover {
130       background: #B91C1C;
131     }
132
133     .btn-outline {
134       background: transparent;
135       border: 2px solid var(--primary);
136       color: var(--text-primary);
137     }
138
139     .btn-outline:hover {
140       background: rgba(15, 118, 110, 0.1);
141     }
142
143     /* Scroll Animations */
144     .reveal {
145       opacity: 0;
146       transform: translateY(30px);
147       transition: all 0.8s cubic-bezier(0.5, 0, 0, 1);
148     }
149
150     .reveal.active {
151       opacity: 1;
152       transform: translateY(0);
```

```
18 class EmergencyPatientManagementSystem:
44     def _compare(self, i, j):
50         p1 = self.heap[i]
51         p2 = self.heap[j]
52         if p1.severity != p2.severity:
53             return p1.severity > p2.severity
54         return p1.arrival_sequence < p2.arrival_sequence
55
56     def _heapify_up(self, i):
57         while i > 0 and self._compare(i, self._parent(i)):
58             self._swap(i, self._parent(i))
59             i = self._parent(i)
60
61     def _heapify_down(self, i):
62         n = len(self.heap)
63         largest = i
64         left = self._left_child(i)
65         right = self._right_child(i)
66
67         if left < n and self._compare(left, largest):
68             largest = left
69         if right < n and self._compare(right, largest):
70             largest = right
71
72         if largest != i:
73             self._swap(i, largest)
74             self._heapify_down(largest)
75
76     # --- Major Operations ---
77
78     def register_patient(self, patient_id, name, severity, doctor=None):
79         """1. Register a new patient and assign priority."""
80         if patient_id in self.patient_map:
81             raise ValueError(f"Patient with ID {patient_id} is already registered.")
82
83         patient = Patient(patient_id, name, severity, doctor)
84         patient.arrival_sequence = self.sequence_counter
85         self.sequence_counter += 1
86
87         # Insert at the end of the heap
88         self.heap.append(patient)
89         index = len(self.heap) - 1
90         self.patient_map[patient_id] = index
91
92         # Restore heap property
93         self._heapify_up(index)
94         print(f"[REGISTERED] {patient}")
95
96     def get_next_patient(self):
97         """2. Identify the next patient for treatment without removing them."""
98         if not self.heap:
99             return None
100         return self.heap[0]
```

```

</> presentation.html 2 X
</> presentation.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>ER Priority System</title>
7      <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800&family=Jetf
8      <style>
9          :root {
10             --primary: #0F766E;
11             --primary-hover: #115E59;
12             --accent: #DC2626;
13             --bg-dark: #0F172A;
14             --bg-card: #1E293B;
15             --bg-surface: #334155;
16             --text-primary: #F8F9FA;
17             --text-muted: #94A3B8;
18             --success: #22C55E;
19             --warning: #F59E0B;
20             --gradient: linear-gradient(135deg, #0F766E, #0EA5E9);
21             --gradient-danger: linear-gradient(135deg, #DC2626, #F59E0B);
22
23             --sev-1: #22C55E;
24             --sev-2: #3B82F6;
25             --sev-3: #F59E0B;
26             --sev-4: #EA580C;
27             --sev-5: #DC2626;
28         }
29
30         * {
31             box-sizing: border-box;
32             margin: 0;
33             padding: 0;
34         }
35
36         body {
37             font-family: 'Inter', sans-serif;
38             background-color: var(--bg-dark);
39             color: var(--text-primary);
40             line-height: 1.6;
41             overflow-x: hidden;
42             scroll-behavior: smooth;
43         }
44
45         /* Typography */
46         h1, h2, h3, h4, h5 {
47             font-weight: 700;
48             line-height: 1.2;
49         }
50
51         h2 {
52             font-size: 2.5rem;
53             margin-bottom: 2rem;
54             text-align: center;

```

```

1  import time
2
3  class Patient:
4      """Represents a patient in the emergency department."""
5      def __init__(self, patient_id, name, severity, doctor=None):
6          self.patient_id = patient_id
7          self.name = name
8          # Higher integer value implies a higher medical severity/priority
9          self.severity = severity
10         self.arrival_time = time.time()
11         # Sequence counter to maintain strict FIFO for patients with identical severity
12         self.arrival_sequence = 0
13         self.doctor = doctor
14
15     def __repr__(self):
16         return f"Patient(ID: {self.patient_id}, Name: {self.name}, Severity: {self.severity}, Doctor: {"
17
18 class EmergencyPatientManagementSystem:
19     """
20     Manages the emergency patient queue using a Max-Heap-based Priority Queue
21     coupled with a Hash Map for O(1) patient lookups.
22     """
23     def __init__(self):
24         self.heap = []
25         self.patient_map = {} # Maps patient_id -> index in the heap array
26         self.sequence_counter = 0
27
28     # --- Heap Helper Methods ---
29     def _parent(self, i):
30         return (i - 1) // 2
31
32     def _left_child(self, i):
33         return 2 * i + 1
34
35     def _right_child(self, i):
36         return 2 * i + 2
37
38     def _swap(self, i, j):
39         self.heap[i], self.heap[j] = self.heap[j], self.heap[i]
40         # Update indices in the hash map after swapping
41         self.patient_map[self.heap[i].patient_id] = i
42         self.patient_map[self.heap[j].patient_id] = j
43
44     def _compare(self, i, j):
45         """
46         Returns True if the patient at index i has HIGHER priority than index j.
47         Priority is primarily decided by severity. If severity is equal,
48         the one who arrived earlier (lower sequence) has higher priority (FIFO).
49         """
50         p1 = self.heap[i]
51         p2 = self.heap[j]
52         if p1.severity != p2.severity:
53             return p1.severity > p2.severity
54         return p1.arrival_sequence < p2.arrival_sequence

```

```

</> presentation.html > ...
 2  <html lang="en">
 3  <head>
 4    <style>
51      h2 {
54        text-align: center;
55        background: var(--text-primary);
56        -webkit-background-clip: text;
57        -webkit-text-fill-color: transparent;
58      }
59
60      code {
61        font-family: 'JetBrains Mono', monospace;
62      }
63
64      /* Layout & Utilities */
65      .container {
66        max-width: 1200px;
67        margin: 0 auto;
68        padding: 0 1.5rem;
69      }
70
71      section {
72        padding: 6rem 0;
73        position: relative;
74      }
75
76      .text-gradient {
77        background: var(--gradient);
78        -webkit-background-clip: text;
79        -webkit-text-fill-color: transparent;
80      }
81
82      .glass-card {
83        background: rgba(30, 41, 59, 0.7);
84        backdrop-filter: blur(10px);
85        border: 1px solid rgba(255, 255, 255, 0.1);
86        border-radius: 1rem;
87        padding: 2rem;
88        box-shadow: 0 10px 30px rgba(0, 0, 0, 0.3);
89        transition: transform 0.3s ease, border-color 0.3s ease;
90      }
91
92      .glass-card:hover {
93        transform: translateY(-5px);
94        border-color: rgba(255, 255, 255, 0.2);
95      }
96
97      .btn {
98        display: inline-flex;
99        align-items: center;
100       justify-content: center;
101       padding: 0.75rem 1.5rem;
102       border-radius: 0.5rem;
103       font-weight: 600;

```

```
18 class EmergencyPatientManagementSystem:
96     def get_next_patient(self):
97         """2. Identify the next patient for treatment without removing them."""
98         if not self.heap:
99             return None
100         return self.heap[0]
101
102     def treat_next_patient(self):
103         """3. Select and remove the highest-priority patient for treatment."""
104         if not self.heap:
105             print("[INFO] No patients waiting in the emergency queue.")
106             return None
107
108         treated_patient = self.heap[0]
109
110         # Remove from hash map
111         del self.patient_map[treated_patient.patient_id]
112
113         last_patient = self.heap.pop()
114         if self.heap:
115             # Move the last patient to the root
116             self.heap[0] = last_patient
117             self.patient_map[last_patient.patient_id] = 0
118             # Restore heap property downwards
119             self._heapify_down(0)
120
121         print(f"[TREATED] {treated_patient}")
122         return treated_patient
123
124     def search_patient(self, patient_id):
125         """4. Search for a particular patient by ID."""
126         index = self.patient_map.get(patient_id)
127         if index is not None:
128             return self.heap[index]
129         return None
130
131     def update_patient(self, patient_id, new_severity=None, new_doctor=None):
132         """5. Update patient information (severity or assigned doctor)."""
133         index = self.patient_map.get(patient_id)
134         if index is None:
135             print(f"[ERROR] Patient ID {patient_id} not found.")
136             return False
137
138         patient = self.heap[index]
139         old_severity = patient.severity
140
141         if new_doctor is not None:
142             patient.doctor = new_doctor
143
144         if new_severity is not None and new_severity != old_severity:
145             patient.severity = new_severity
146             # Adjust the patient's position in the heap based on new priority
147             if new_severity > old_severity:
148                 self._heapify_up(index)
```

</> index.html X

</> index.html > ...

```
2   <html lang="en">
3   <head>
13     <style>
47       body {
54       }
55
56       ::-webkit-scrollbar { width: 8px; height: 8px; }
57       ::-webkit-scrollbar-track { background: transparent; }
58       ::-webkit-scrollbar-thumb { background: #C6C6C6; border-radius: 4px; }
59       ::-webkit-scrollbar-thumb:hover { background: #A8A8A8; }
60
61       /* Sidebar */
62       .sidebar {
63         width: 260px;
64         background: var(--bg-card);
65         border-right: 1px solid var(--border-solid);
66         padding: 24px 16px;
67         display: flex;
68         flex-direction: column;
69         z-index: 50;
70       }
71
72       .brand {
73         display: flex;
74         align-items: center;
75         gap: 12px;
76         font-size: 22px;
77         font-weight: 800;
78         color: var(--text-dark);
79         margin-bottom: 40px;
80         padding: 0 12px;
81         letter-spacing: -0.5px;
82       }
83       .brand i { color: var(--brand-main); }
84
85       .nav-item {
86         display: flex;
87         align-items: center;
88         gap: 12px;
89         padding: 12px 16px;
90         margin-bottom: 4px;
91         border-radius: var(--radius-sm);
92         color: var(--text-muted);
93         text-decoration: none;
94         font-weight: 600;
95         font-size: 15px;
96         transition: all 0.2s ease;
97       }
98       .nav-item i { width: 18px; height: 18px; stroke-width: 2.5px; }
99       .nav-item:hover { background: var(--border-light); color: var(--text-dark); }
100      .nav-item.active { background: var(--brand-light); color: var(--brand-main); }
101
102      .admin-badge {
103        margin-top: auto;
```

</> index.html ×

</> index.html > ...

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>ER Command Center</title>
7   <!-- Premium Font -->
8   <link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;700;800&fa"
9   <!-- Supabase, ChartJS, Lucide -->
10  <script src="https://cdn.jsdelivr.net/npm/@supabase/supabase-js@2"></script>
11  <script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.1/dist/chart.umd.min.js"></script>
12  <script src="https://unpkg.com/lucide@latest"></script>
13  <style>
14    :root {
15      --brand-main: #0F62FE; /* Crisp professional blue */
16      --brand-hover: #0043CE;
17      --brand-light: #EDF5FF;
18
19      --bg-body: #F4F4F4;
20      --bg-card: #FFFFFF;
21
22      --text-dark: #161616;
23      --text-muted: #525252;
24      --text-light: #8D8D8D;
25
26      --border-solid: #E0E0E0;
27      --border-light: #F4F4F4;
28
29      /* Enhanced Severity Palette */
30      --sev-1: #198038; --sev-1-bg: #DEFBE6;
31      --sev-2: #0F62FE; --sev-2-bg: #EDF5FF;
32      --sev-3: #F1C21B; --sev-3-bg: #FCF4D6; --sev-3-text: #8A6D00;
33      --sev-4: #FF832B; --sev-4-bg: #FFF1E8;
34      --sev-5: #DA1E28; --sev-5-bg: #FFF1F1;
35
36      --shadow-sm: 0 2px 4px rgba(0,0,0,0.04);
37      --shadow-md: 0 4px 8px rgba(0,0,0,0.04), 0 0 2px rgba(0,0,0,0.06);
38      --shadow-lg: 0 12px 24px rgba(0,0,0,0.06), 0 0 4px rgba(0,0,0,0.04);
39
40      --radius-lg: 16px;
41      --radius-md: 12px;
42      --radius-sm: 8px;
43    }
44
45    * { box-sizing: border-box; margin: 0; padding: 0; font-family: 'Plus Jakarta Sans', sans-serif; }
46
47    body {
48      background-color: var(--bg-body);
49      color: var(--text-dark);
50      display: flex;
51      height: 100vh;
52      overflow: hidden;
53      -webkit-font-smoothing: antialiased;
```

Medical Staff

 Search patients by name or ID...



Dr. Emily Chen

Cardiology

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Avail

Busy

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Avail

Busy

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Avail

Busy

Dr. Sarah Wilson

Emergency Medicine

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 sarah.wilson@hospital.com

Avail

Busy

Triage Queue

Search patients by name or ID...



8 WAITING 2 CRITICAL 2 SEVERE

TREAT HIGHEST PRIORITY

P002 Jane Smith LEVEL 5 - CRITICAL

32y F Unassigned

CHIEF COMPLAINT
Chest pain, difficulty breathing

Arrived 1h ago Status: WAITING

Change Triage Start Treatment

P008 Fatima Hassan LEVEL 5 - CRITICAL

38y F Unassigned

CHIEF COMPLAINT
Severe allergic reaction, swelling

Arrived 1h ago Status: WAITING

Change Triage Start Treatment

Register Patient

Search patients by name or ID...



PATIENT INFORMATION

Patient ID (Auto)

P009

Full Name *

John Doe

Age

e.g. 45

Gender

Male

Contact Phone

+1 ...

VITALS (OPTIONAL)

Blood Pressure

120/80

mmHg

Heart Rate

80

bpm

O2 Sat

98

%

Temp

98.6

°F

TRIAGE ASSESSMENT *

Select Severity Level

1

2

3

4

5